



Operating System: Basics

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Characteristics of Modern OS

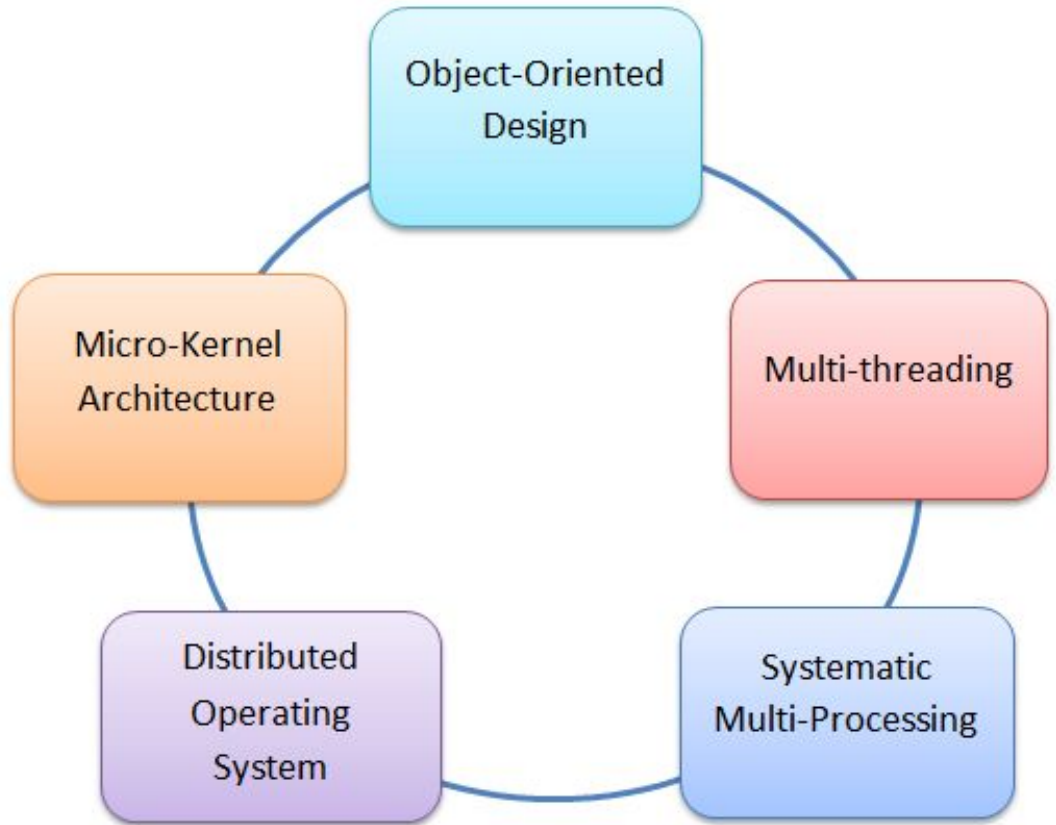


Fig: Characteristics of Modern Operating System

Object-Oriented Design

- An object-oriented operating system is an operating system that uses object-oriented design principles.
- Typical operating systems organize their data in "files". An object-oriented system will talk in objects instead, each of them being instances of a class.
- current operating systems are, in a way, procedure-oriented. They usually operate on raw binary data; object-oriented extensions are only implemented in individual programs.
- a file, its format, and the code which deals with the format all belong together, and OOSys honors this.
- Advantage
 - By describing all elements of software with one single concept, users only need to learn how to deal with this single concept; after that, they can quickly adapt to new elements.
 - Users can see immediately how their operations affect the objects they work on.
 - OOSys can provide cross-platform compatibility, which then includes all of its classes.
 - Code does not need to be compiled. program code is interpreted and not compiled and optimized. Programmers can reuse existing code more easily.
- Restrictions
 - Speed-critical systems cannot be ported to OOSys, as code written in the OOSys language is interpreted and therefore slower than usual compiled code.
 - OOSys will never be able to guarantee security and stability. If it is implemented as free software, there will not be any possible warranty either.

Multi-threading

- An application or program is divided into several smaller tasks, these tasks are executed by the processor concurrently.
- These smaller tasks are called threads. A thread is a lightweight process or a dispatchable unit of work.
- A thread has some of the characteristics of a process. but the difference is several threads sharing the same memory space, the process doesn't share the memory space.



One Process
Multiple Threads



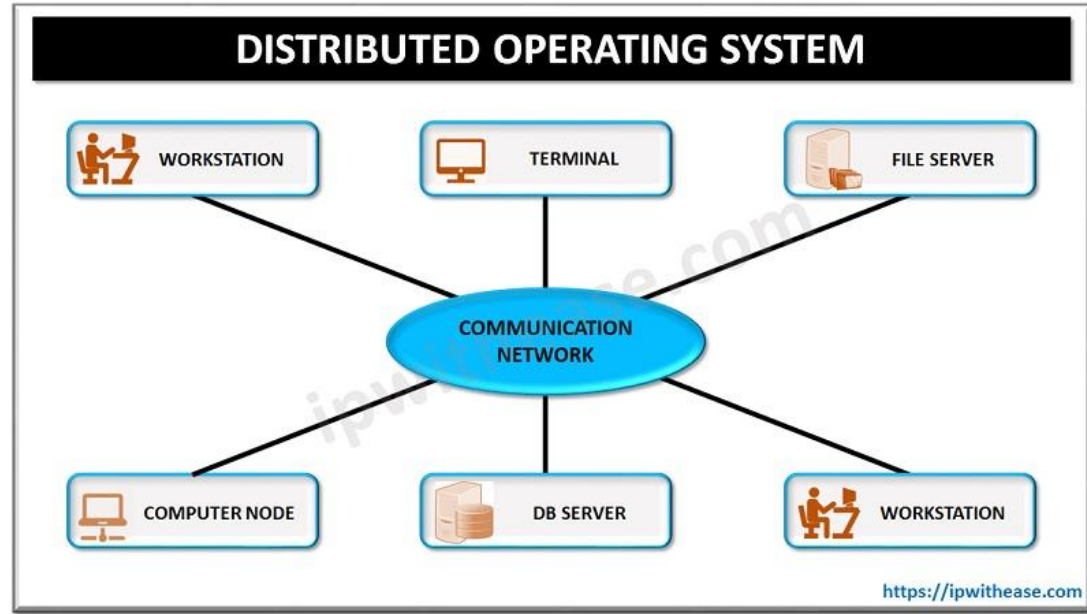
Multiple Process Multiple
Threads per Process

Systematic Multi-Processing

- In a Symmetrical multiprocessing system, each processor executes the same copy of the operating system, takes its own decisions, and cooperates with other processes to smooth the entire functioning of the system.
- a "shared every-thing" system, because the processors share memory and the Input output bus or data path.
- Any new job submitted by a user can be assigned to any processor that is least burdened.
- In this system, any processor can run any job or process.
- In this, any processor initiates an Input and Output operation.
- These systems are fault-tolerant. Failure of a few processors does not bring the entire system to a halt.
- It is very difficult to balance the workload among processors rationally.
- Asymmetric multiprocessing operating system -
 - There is a master slave relationship between the processors.
 - one processor may act as a master processor or supervisor processor while others are treated as slave

Distributed Operating System

- Distributed Operating System (DOS) is a model which is designed to run distributed applications on multiple computer systems efficiently combined by communications.
- In this operating system several constituents interact with each other to perform actions that seems as a single working system to the end user.
- In distributed operating system the most salient feature is sharing of resources and this sharing is managed by servers.



- **Resource Sharing**
- **Security:** The security is maintained in a way that client sends request for the access on resources. The server authenticates the user and then after authentication the server grants access
- **Fault Reconginition:**
- **Fault Tolerance:**
- **Scalability:** how the system handles if there will be a growth in the number of users and continuous integration on system.
- **Heterogeneity:** The system was composed of several programming languages, operating systems , software, hardware,

Types of Distributed Operating System

- **Client-Server:**

- In client-server, only server have the rights to share, allocate or de-allocate the shared resources to clients on need basis.
- The client sends a request to server for resource, and then the server provides the resources which were requested by the client.
- Both the client and server interact with each other with the communication network in DOS.

- **Peer to Peer(P2P):**

- All connected devices and systems can act as client and as well as a server too.
- All the nodes are equal participants in sharing resources.
- The devices interact with each other through the communication network for sharing resources with each other on need basis.

- **N-tier:**

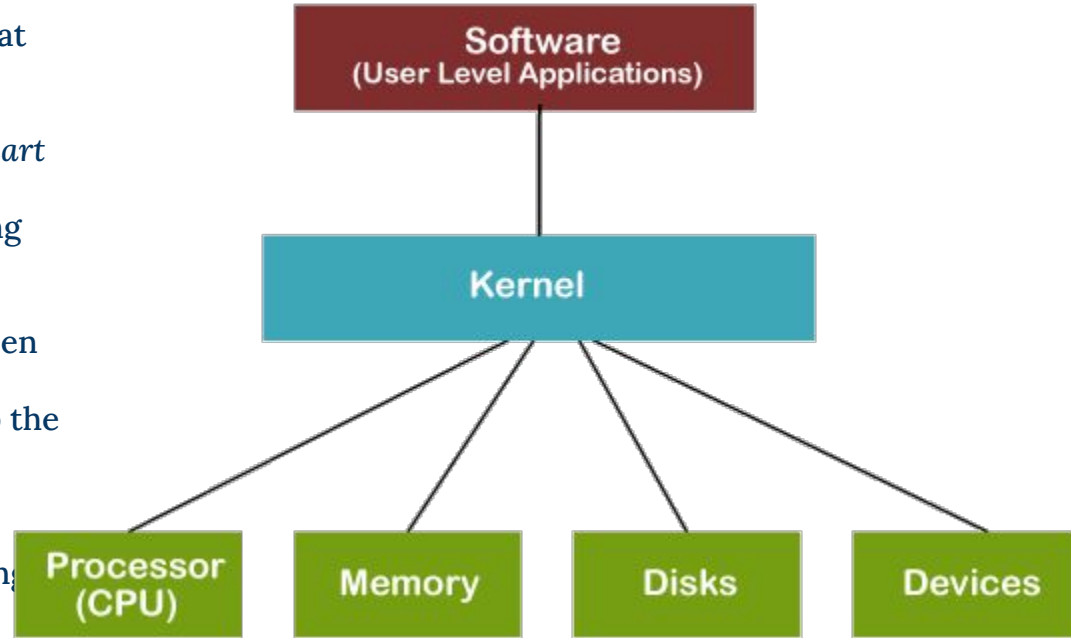
- When we need to forward the server or applications to other additional organization or company on the network than we will use **n-tier**.

- **Three-tier:**

- In DOS, three tier is composed of three layers i.e. presentation tier, data storage tier and a domain logic tier.
- This type provides reliability and flexibility to development teams that allow them to work on particular piece of an application separately.

Micro-Kernel Architecture: What is kernel

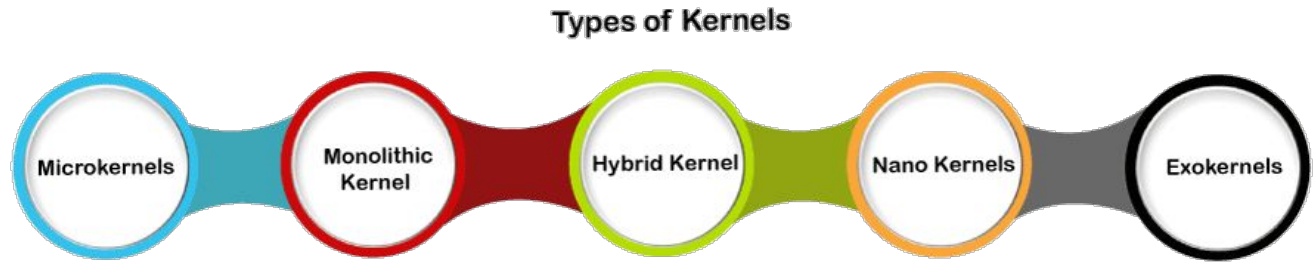
- A microkernel is a minimal operating system that performs only the essential functions of an operating system.
- Kernel is a computer program that is a core or heart of an operating system.
- Bridge between applications and data processing done at the hardware level.
- Part of the OS that always resides in computer memory and enables the communication between software and hardware components.
- computer program that first loaded on start-up the system (After the bootloader).
- It also manages memory, peripheral, and I/O requests from software.
- It translates all I/O requests into data processing instructions for the CPU.
- It manages other tasks also such as memory management, task management, and disk management.



Kernel Contd:

- A kernel is kept and usually loaded into separate memory space, known as protected Kernel space.
- Other application programs such as browser, word processor, audio & video player use separate memory space known as user-space.
- Due to these two separate spaces, user data and kernel data don't interfere with each other
- **Functions of a Kernel**
 - A kernel is responsible for controlling I/O devices using device drivers. Here, a device driver is a computer program that helps or enables the OS to communicate with any hardware device.
 - The kernel has full control for accessing the computer's memory.
 - To allocate the memory, the first step is known as **virtual addressing**, which is done by paging or segmentation. **Virtual addressing** is a process of providing virtual address spaces to the processes.
 - Share the resources between various processes. It must share the resources in a way that each process uniformly accesses the resource.
 - The kernel also provides a way for synchronization and **inter-process communication (IPC)**. It is responsible for context switching between processes.
 - Kernel decides which memory each process will use and what to do if the required memory is not available.

Kernel: Type

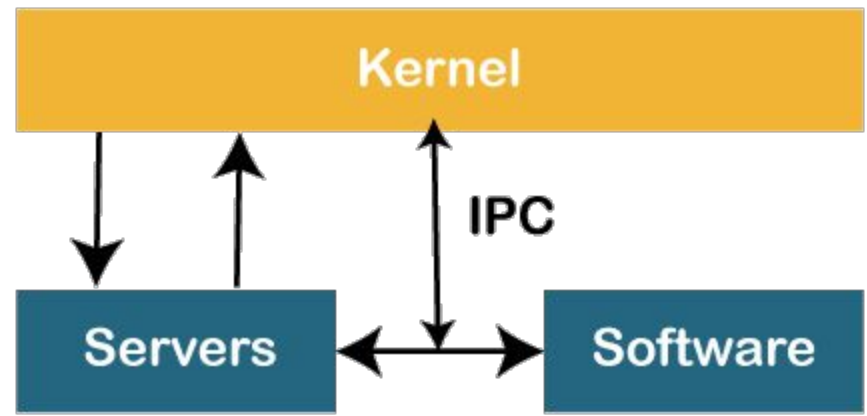


- **Monolithic Kernels**

- **same memory space is used to implement user services and kernel services.**
- there is no different memory used for user services and kernel services.
- As it uses the same memory space, the size of the kernel increases, increasing the overall size of the OS.
- **Examples - Unix, Linux, Open VMS, XTS-400, etc.**
- **Advantages -**
 - The execution of processes is also faster as there is no separate user space and kernel space and less software involved.
 - As it is a single piece of software hence, it's both sources and compiled forms are smaller
- **Disadvantages:**
 - If any service generates any error, it may crash down the whole system.
 - These kernels are not portable, which means for each new architecture, they must be rewritten.
 - To add a new service, the complete operating system needs to be modified.

Microkernel

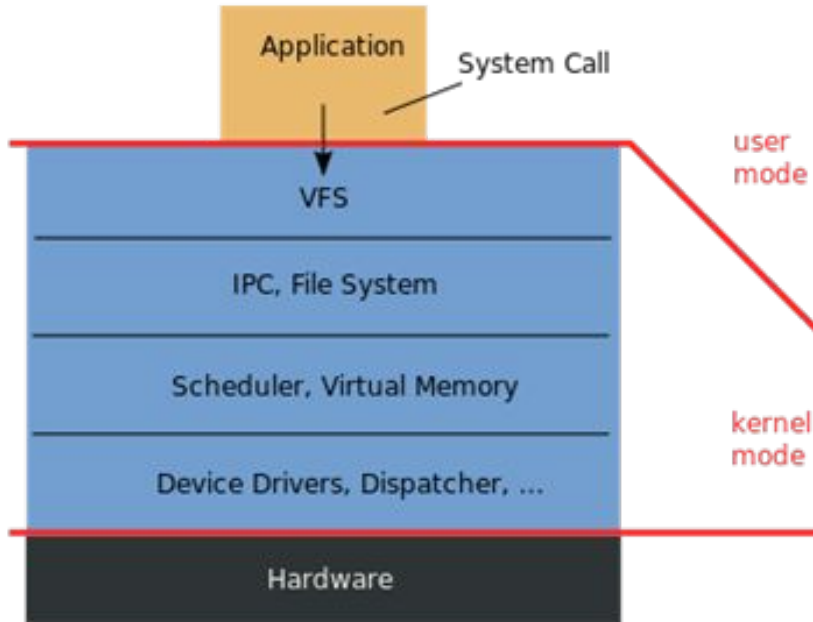
- User services and kernel services are implemented into two different address spaces: user space and kernel space.
- Microkernels are easier to manage and maintain
- There will be a greater number of system calls and context switching
- Use a message passing system for handling the request from one server to another server.
- Only some essential services are provided by microkernels, such as defining memory address spaces, Interprocess Communication, and process management.
- Other services such as networking are not provided by Kernel and handled by a user-space program known as servers.
- If a kernel process crashes, the crashing of the whole system can still be prevented
- Examples - L4, AmigaOS, Minix, K42, etc.



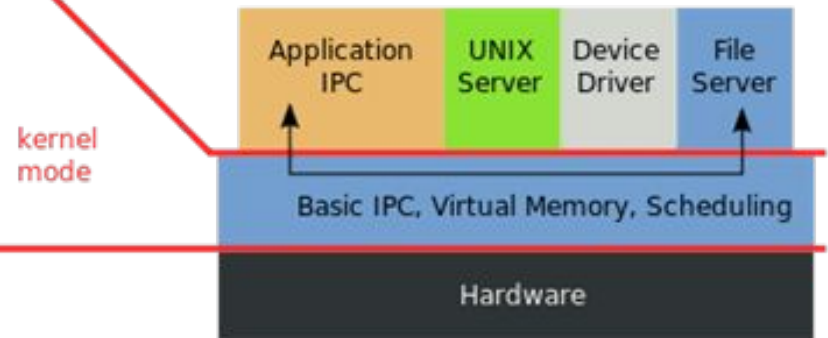
- Microkernels can be managed easily.
- A new service can be easily added without modifying the whole OS.
- In a microkernel, if a kernel process crashes, it is still possible to prevent the whole system from crashing.
- There is more requirement of software for interfacing, which reduces the system performance.
- Process management is very complicated.
- The messaging bugs are difficult to fix.

Comparison

Monolithic Kernel based Operating System

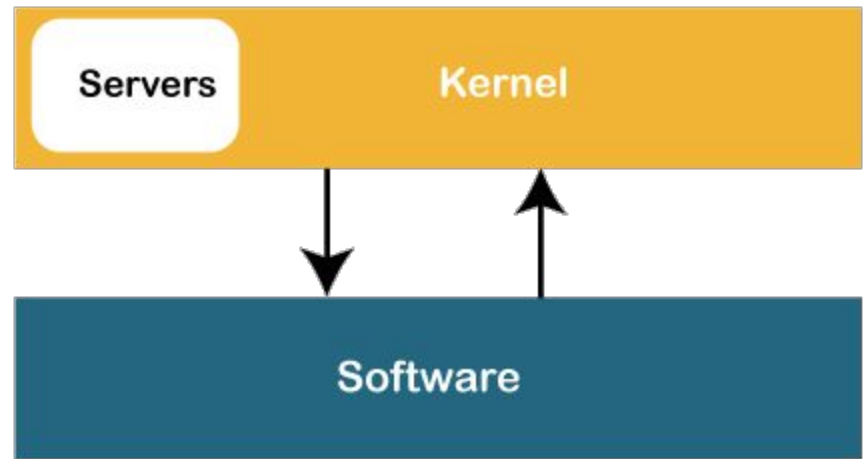


Microkernel based Operating System



Hybrid Kernel

- Modular kernels
- Combination of both Monolithic and Microkernels.
- Speed of monolithic kernels
- Modularity of microkernels.
- These kernels are widely used in commercial OS,
 - Such as different versions of MS Windows.



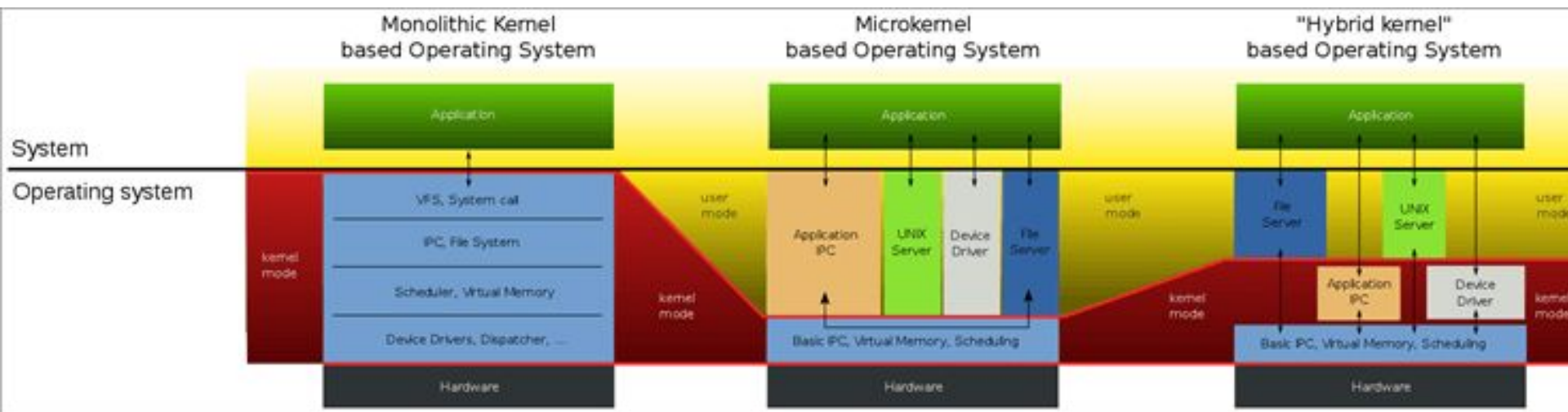
Advantages:

- There is no requirement for a reboot for testing.
- Third-party technology can be integrated rapidly.

Disadvantages:

- There is a possibility of more bugs with more interfaces to pass through.
- It can be a confusing task to maintain the modules for some administrators, especially when dealing with issues such as symbol differences.

Comparison

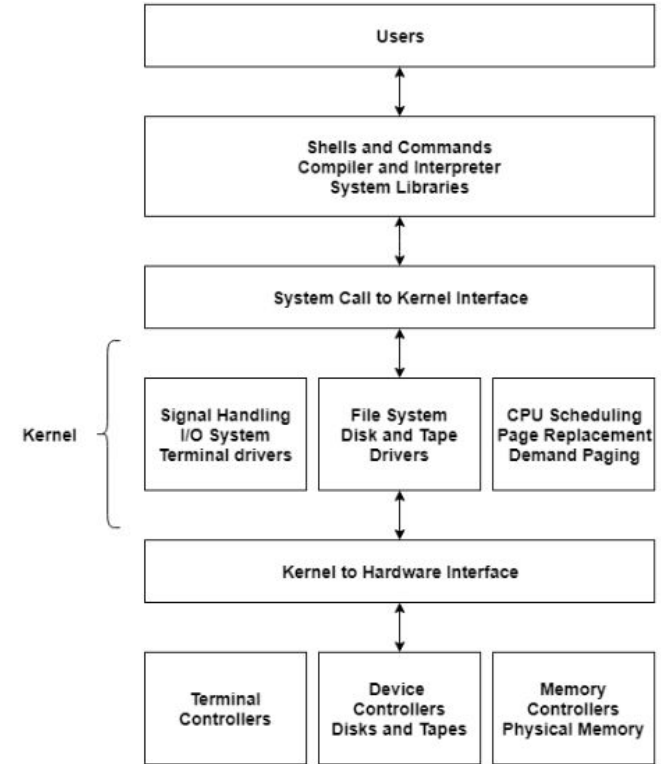


Nanokernel and Exokernel

- *Nanokernel - the complete code of the kernel is very small, which means the code executing in the privileged mode of the hardware is very small.*
 - Examples - EROS
- Adv - It provides hardware abstractions even with a very small size.
- Issue - Nanokernel lacks system services. Not in much practical use
- Exokernel - Experimental approach for designing OS.
- Resource protection is kept separated from management, which allows us to perform application-specific customization.
- Adv - can incorporate multiple library operating systems. Each library exports a different API, such as one can be used for high-level UI development, and the other can be used for real-time control.
- Issue - The design of the exokernel is very complex.

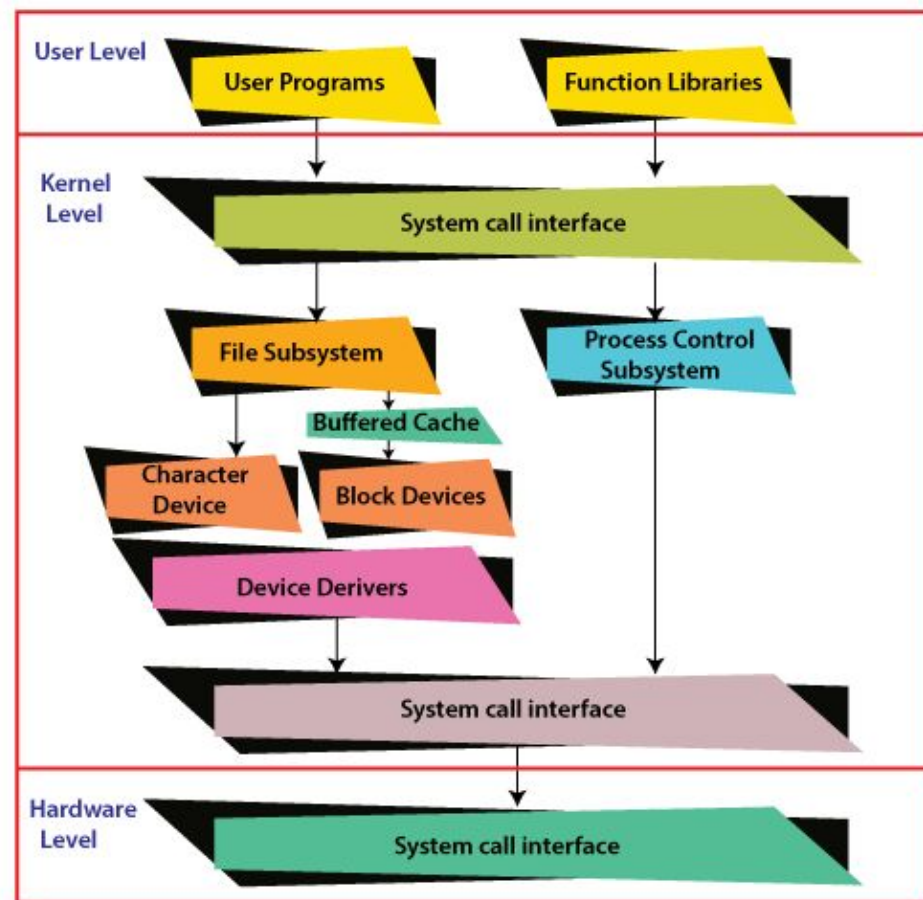
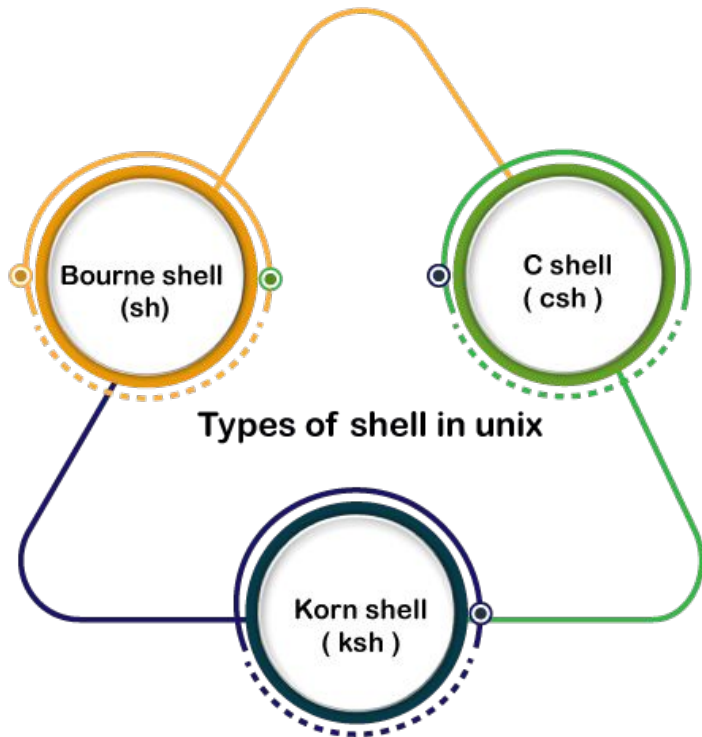
Structure of a UNIX System

- Unix is a multiuser, multitasking operating system
- The kernel provides a bridge between the hardware and the user. handles the files, memory, devices, processes and the network for the operating system. It is the responsibility of the kernel to make sure all the system and user tasks are performed correctly.
- The program between the user and the kernel is known as the shell. It translates the many commands that are typed into the terminal session.
- The applications and utility layer in Unix includes the word processors, graphics programs, database management programs, commands etc. The application programs provide an application to the end users.



Unix Operating System Structure

Structure of Unix System



Kernel Architecture

Difference Between Unix and Linux

	Linux	Unix
OS	Unix-like, open course	Unix, Mixed
Kernel	Monolithic kernel	Kernel Type varies. It can be monolithic, microkernel and hybrid.
Ex -	Debian, Ubuntu, Fedora, Red Hat, Android, etc.	IBM AIX, Solaris, HP-UX, Darwin, macOS X, etc.
Threat	As Linux is mainly driven by open source community, many developers across different parts of the world are working on the code. Hence threat detection and solution is quite fast in case of Linux.	Due to the proprietary nature of Unix, users need to wait for proper bug fixing patches.
Price	Linux is free. However, corporate support is available at a price.	Unix is not free. However, some Unix versions are free for development use (Solaris).
Written	C and other programming languages.	C and assembly language.
features	Provides a great feature through which the kernel components like device drives can be dynamically added and removed as modules.	Traditional Unix systems kernel need static linking of new systems being added